

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	25 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Applicant Data Management	Enter applicant personal details Enter income details Enter loan details Store applicant information
FR-2	Data Preprocessing	Handle missing values Encode categorical data Feature scaling and transformation
FR-3	Loan Eligibility Prediction	Apply Decision Tree model Apply Random Forest model Apply KNN model Apply XGBoost model
FR-4	Prediction Result Management	Display loan approval/rejection result Show prediction confidence score Store prediction history
FR-5	Data Visualization & Analysis	Generate count plots Generate distribution plots Generate bar charts Analyze loan approval trends
FR-6	Web Application Interface	Home page Prediction page User input forms Result display page
FR-7	Model Management	Train machine learning models Evaluate model performance Select best model (XGBoost) Save model using Pickle
FR-8	Deployment & Access	Deploy application on IBM Cloud Access through web browser Provide real-time predictions

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Simple and user-friendly interface for loan applicants and credit officers
NFR-2	Security	Secure handling of applicant information and input validation
NFR-3	Reliability	Consistent and accurate loan eligibility predictions
NFR-4	Performance	Prediction results should be generated within a few seconds
NFR-5	Availability	System should be accessible through web browsers at all times
NFR-6	Scalability	Able to handle increasing numbers of loan applications and users
NFR-7	Maintainability	Easy model updates and application maintenance
NFR-8	Accuracy	Machine learning model should provide high prediction accuracy and minimize false approvals/rejections